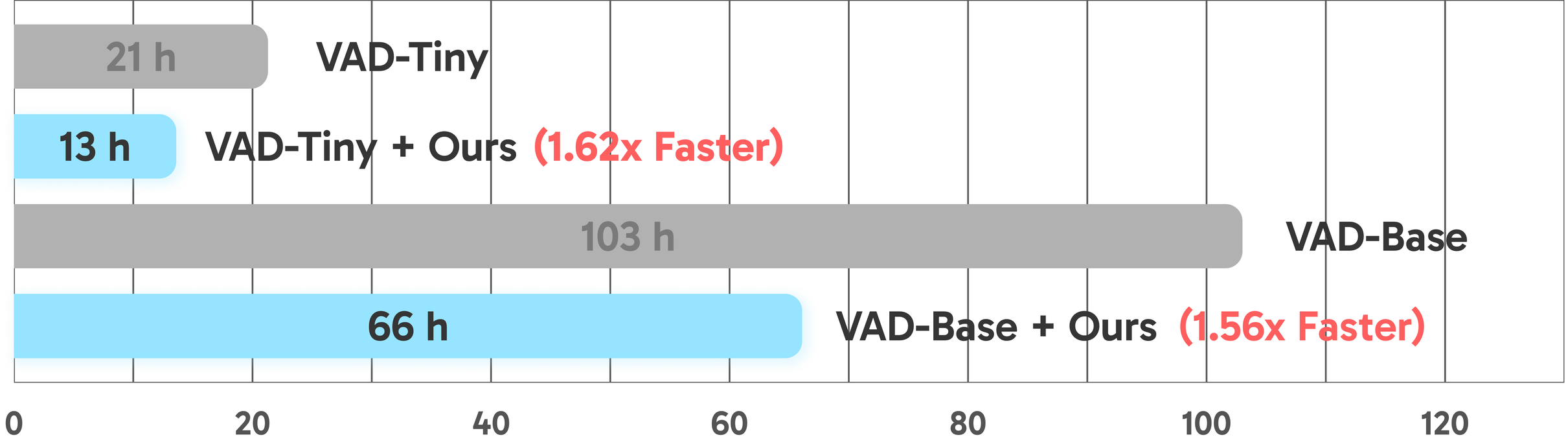
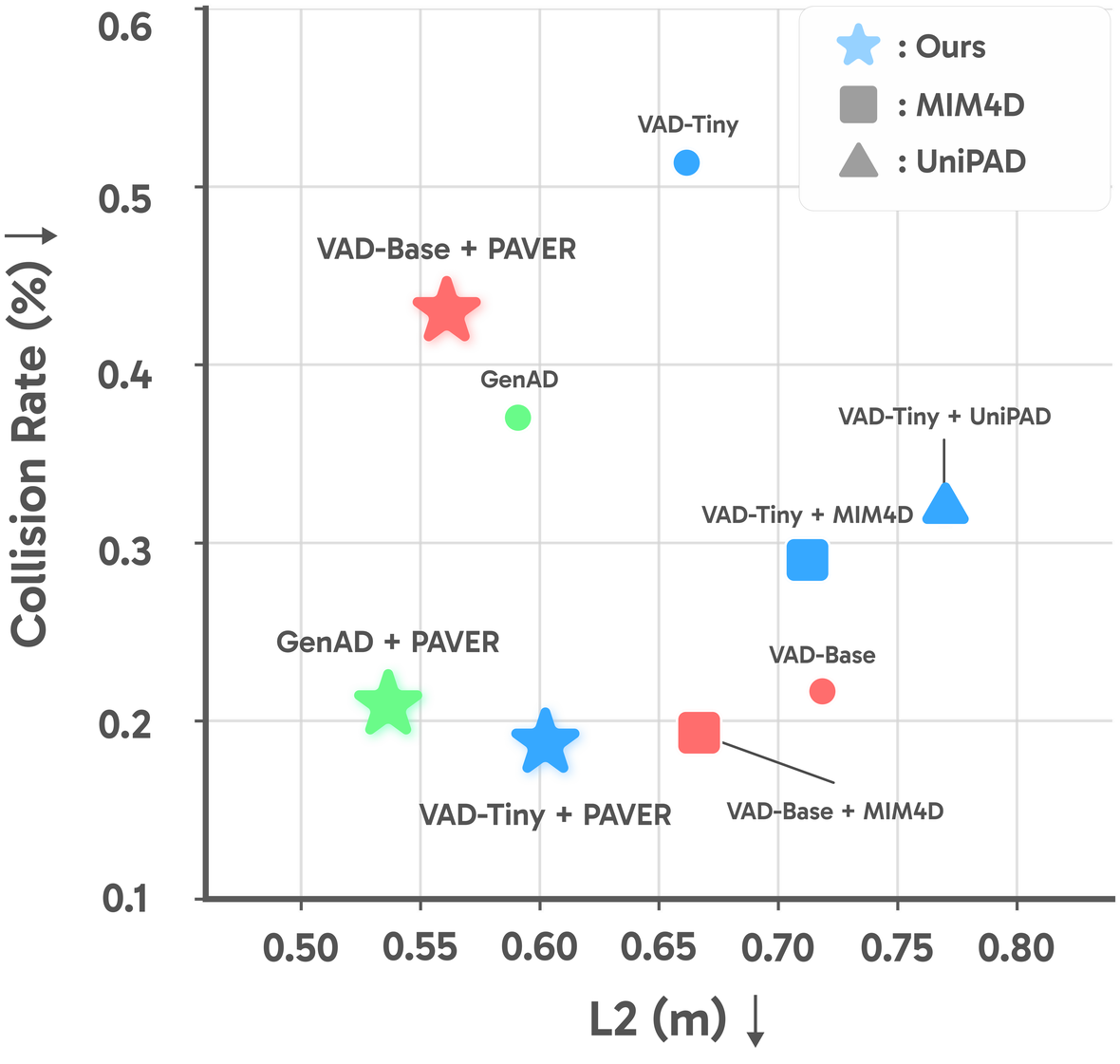
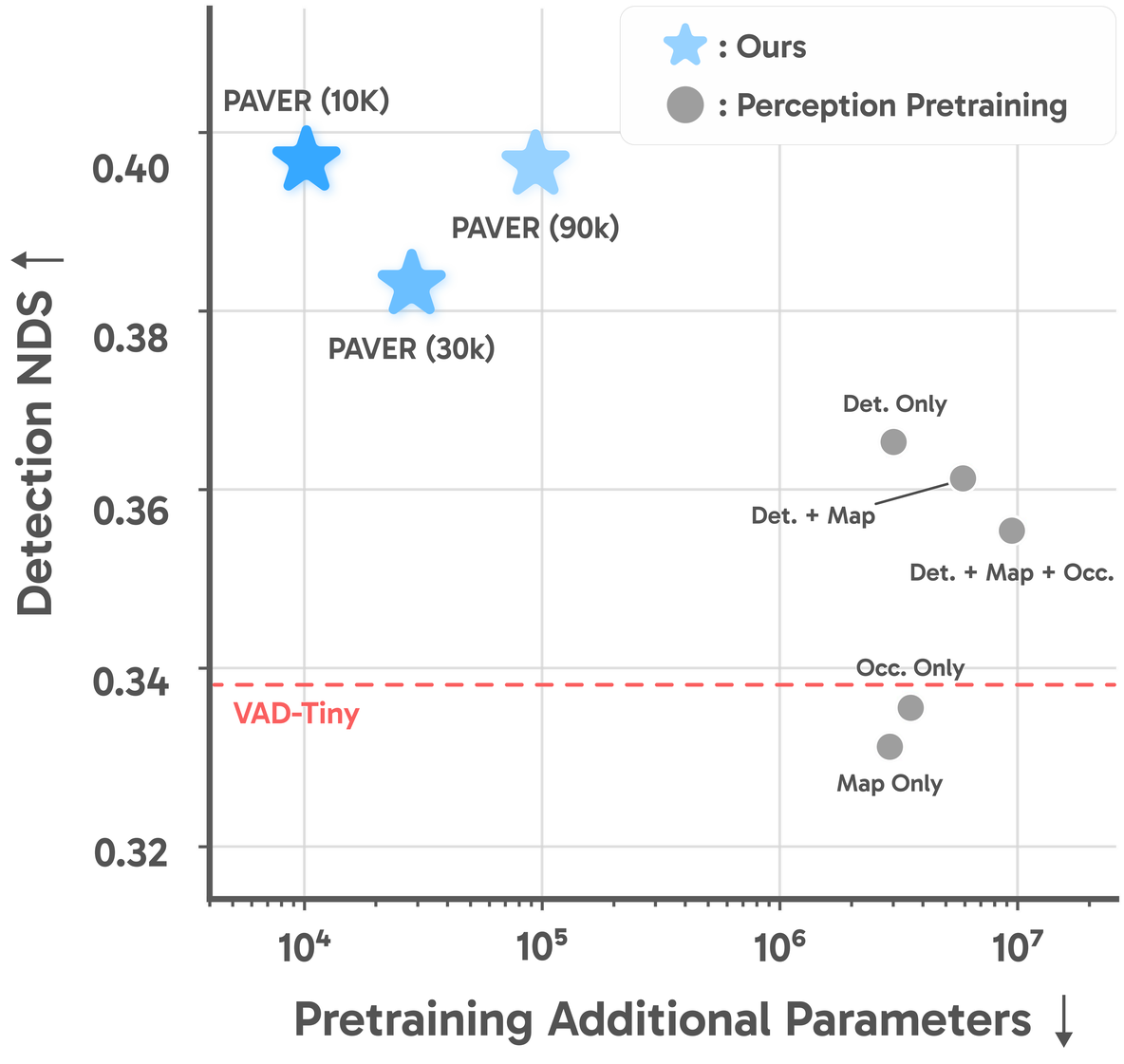

21 h
VAD-Tiny
13 h
VAD-Tiny + Ours (1.62x Faster)
103 h
VAD-Base
66 h
VAD-Base + Ours (1.56x Faster)
0
20
40
60
80
100
120
Total Training Time (h)


: Ours
: MIM4D
: UniPAD
VAD-Tiny
VAD-Base + PAVER
GenAD
VAD-Tiny + UniPAD
VAD-Tiny + MIM4D
GenAD + PAVER
VAD-Base
VAD-Tiny + PAVER
VAD-Base + MIM4D
0.6
0.5
0.4
0.3
0.2
0.1
Collision Rate (%)
0.50
0.55
0.60
0.65
0.70
0.75
0.80
L2 (m)


: Ours
: Perception Pretraining
PAVER (10K)
PAVER (90k)
PAVER (30k)
Det. Only
Det. + Map
Det. + Map + Occ.
Occ. Only
VAD-Tiny
Map Only
0.40
0.38
0.36
0.34
0.32
Detection NDS
10^4
10^5
10^6
10^7
Pretraining Additional Parameters